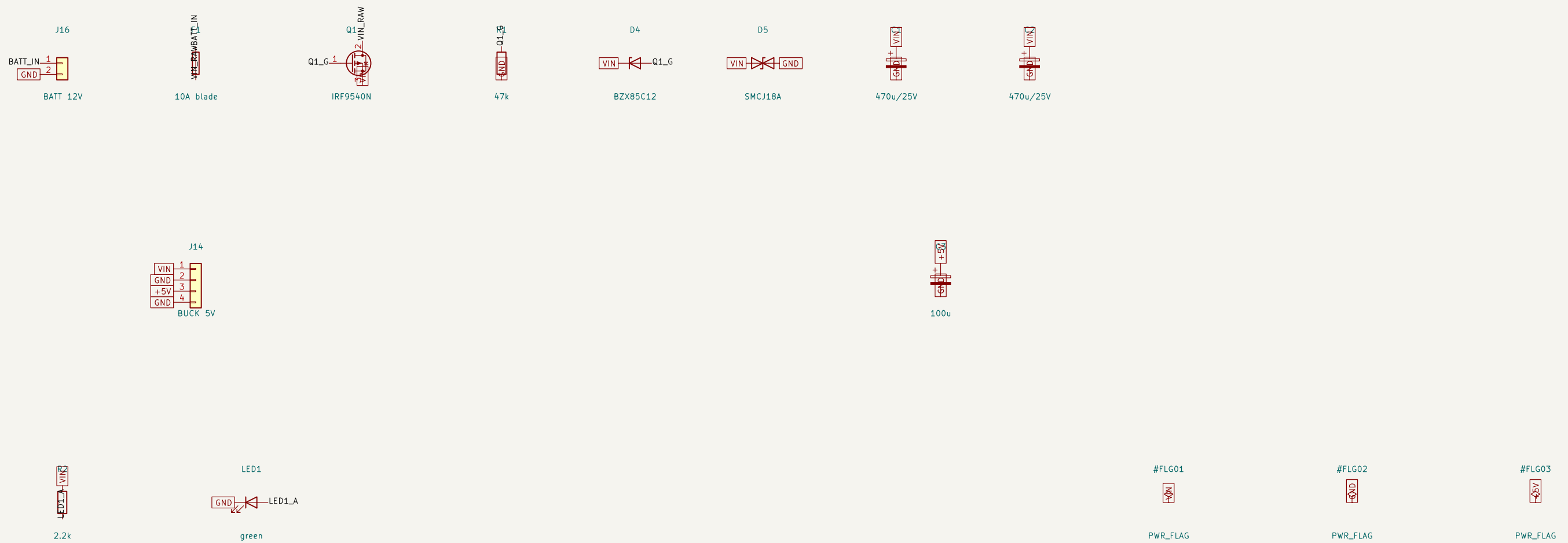




POWER: 12V battery in, protection, generic 5V buck on J14

J14: VIN GND 5V GND – wire to an XL4015/LM2596–class module, set 5.1V BEFORE fitting

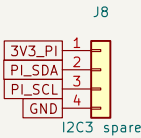
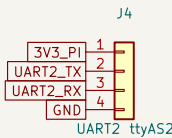
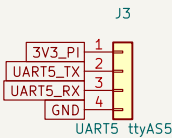
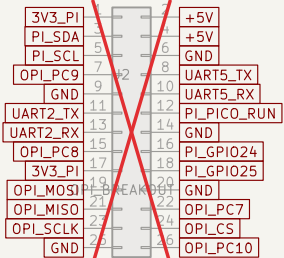
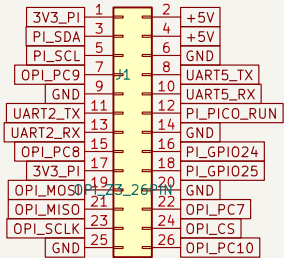
Servo bridge runs on protected VIN directly (12V-only board)



ORANGE PI ZERO 3 CARRIER: 26-pin socket (module on top), UART breakouts

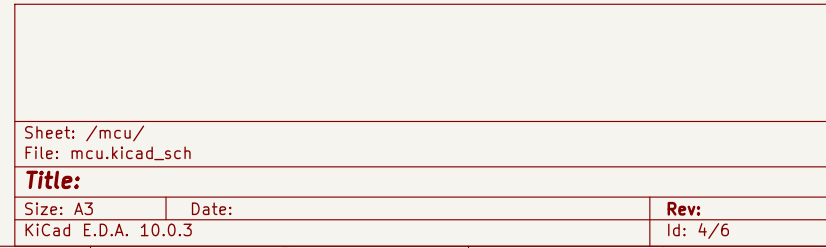
Module is powered through header 5V pins – do NOT also plug USB-C power

UART TX/RX directions per OPI manual – if a device stays silent, swap TX/RX at the JST



PWR_FLAG

Pico soldered directly (low profile, sits under the Pi). Pi reflashes it via SWD on GPIO24/25 (openocd).





SERVO: on-board 2x BTN8982TA half-bridge (BTS7960-compatible interface), VM = +12V rail
AS5600 (in servo housing per vanchor-ng cad/) cabled to J11; keep lead <1m, 100kHz
24-48V boats: servo stall limited by BUCK2 (4.5A). 12V boats: J17 links VIN->12V, full stall current

